

Midterm 2 instructor guide

Slide route: Syllabus date: November 18, 2026. Proposed coverage: Lectures 5–7, with earlier prerequisites.

Session 10 | Midterm 2

Purpose: Second assessment session; no new content. The syllabus gives November 18. Calendar placement and exam duration require confirmation against the actual timetable.

Proposed blueprint

Area	Share	Evidence of learning
Multiple regression	35%	OVB sign, partial interpretation, reference categories.
Joint inference	30%	Joint null, restrictions, covariance-aware test.
Functional forms	35%	Log units, quadratic effect, interaction contrast.

The blueprint is a planning aid, not an approved exam. Announce final coverage in advance. If the course requires cumulative coverage for this midterm, adjust the blueprint explicitly.

Step-by-step exam-day script

Step 1 | Arrival and instructions

Guide Confirm the official paper, permitted materials, timing, and accessibility arrangements before students begin.

Step 2 | OVB question

Guide Require both links in the sign argument: the omitted variable affects the outcome and is correlated with the included regressor.

Step 3 | Multiple regression

Guide Ask for a coefficient interpretation with the conditioning phrase “holding other included regressors fixed.”

Step 4 | Joint test

Guide Make students write the joint null and count restrictions before using an F or Wald statistic.

Step 5 | Equality restriction

Guide Test $\beta_1 = \beta_2$ as one restriction and include the covariance in the standard error.

Step 6 | Functional form

Guide Give a quadratic or log coefficient and require units, comparison change, and exact-versus-approximate language.

Step 7 | Interaction

Guide Ask for the group contrast at a stated X value, not just the main dummy coefficient.

Step 8 | Interpretation check

Guide Include one tempting but false causal conclusion and award credit for identifying the missing assumption.

Step 9 | Review and submission

Guide Give a final warning, collect all pages, and note ambiguous wording for a post-exam diagnostic.

100-minute slot assumption Ten minutes for instructions, 80 for the paper, ten for collection; replace with official timing if different. Confirm approved resources, room, and accessibility arrangements in advance.

What students should know A regression coefficient needs units and a conditioning statement. A joint test needs a joint null. An interaction effect needs a specified value of the interacting variable.

Instructor calibration examples

OVB An omitted positive determinant of Y negatively correlated with X creates negative bias in a simple-regression slope under the usual linear setup. Award credit for both links in the argument.

Restrictions Testing $\beta_1 = \beta_2$ is one restriction, not two. The variance of their estimated difference includes their covariance.

Interactions In $Y = 5 + 2X + 3D + 4XD + u$, the group contrast at $X = 2$ is $3 + 8 = 11$. The slope difference is 4. These answer different questions.

Logs A log-outcome coefficient .1 means an approximate 10% change, with exact exponentiated-index change $100(e^{.1} - 1) \simeq 10.52\%$.

After the paper

Record which errors concern algebra and which concern interpretation. Prepare a short return-to-teaching activity using a strong regression result with a doubtful causal story. This provides the transition to threats to identification.

Next session Lecture8v2.tex. Ask: even with correct standard errors and a flexible functional form, what can still go wrong?